

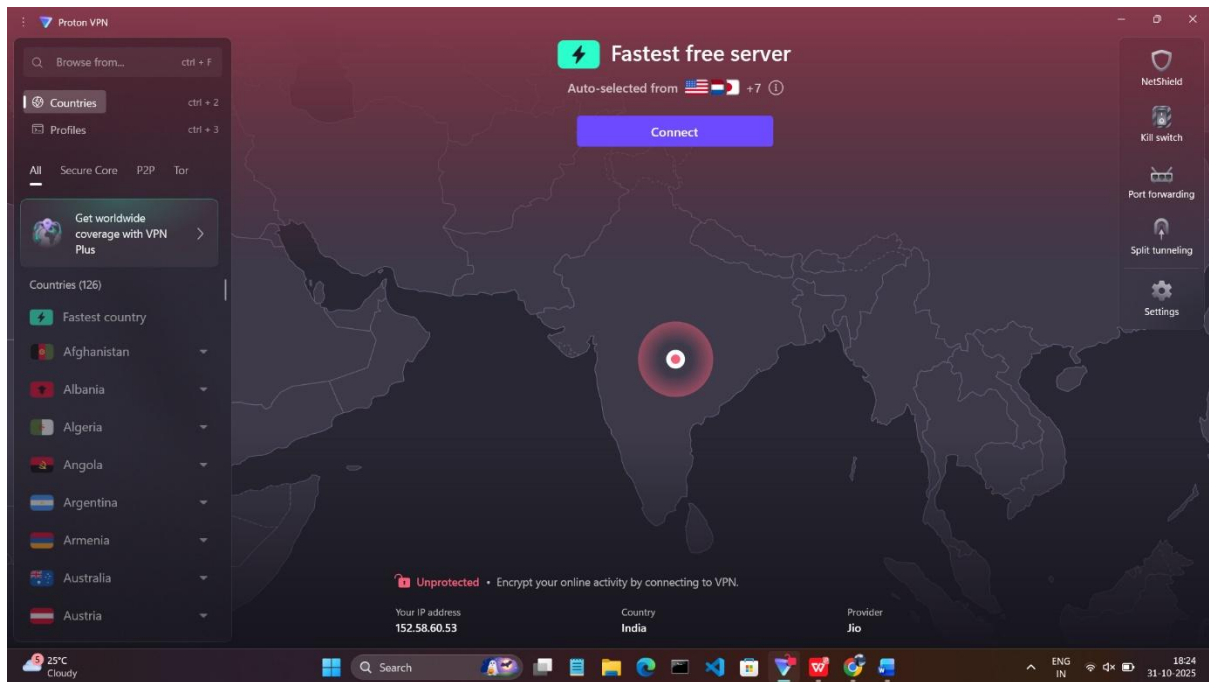
Task 8: Working with VPNs.

Aim: Understand the role of VPNs in protecting privacy and secure communication.

Requirements: Free VPN client - ProtonVPN.

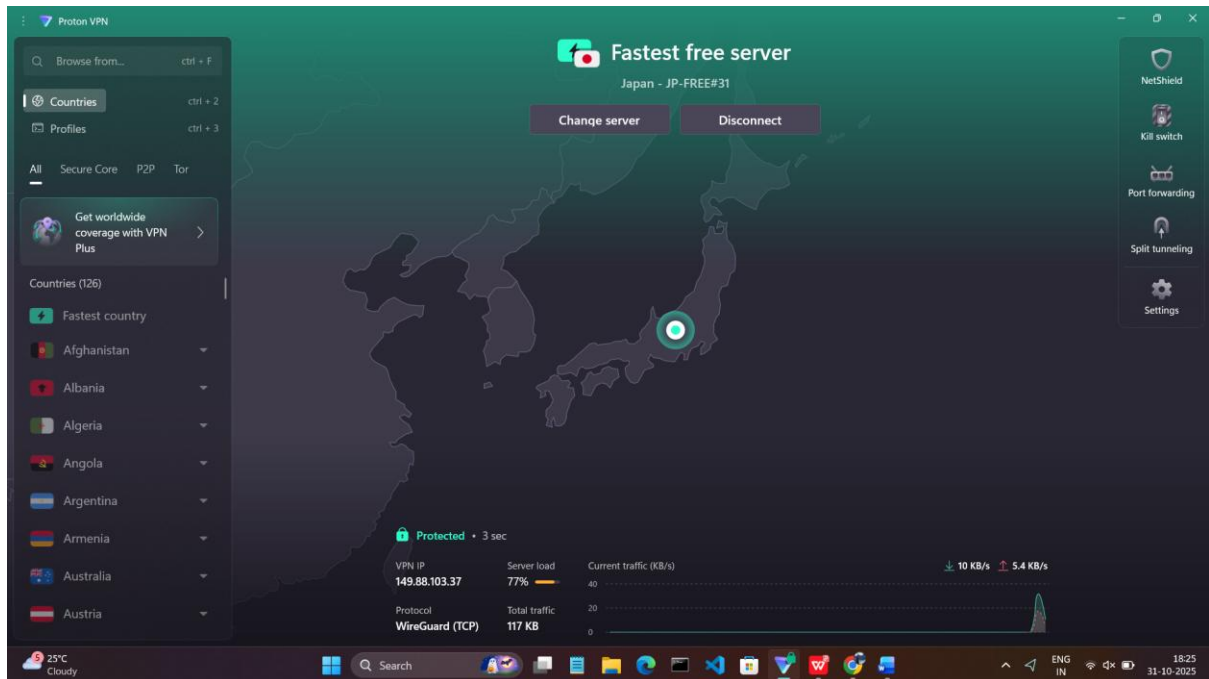
Steps:

1. Open the VPN app.

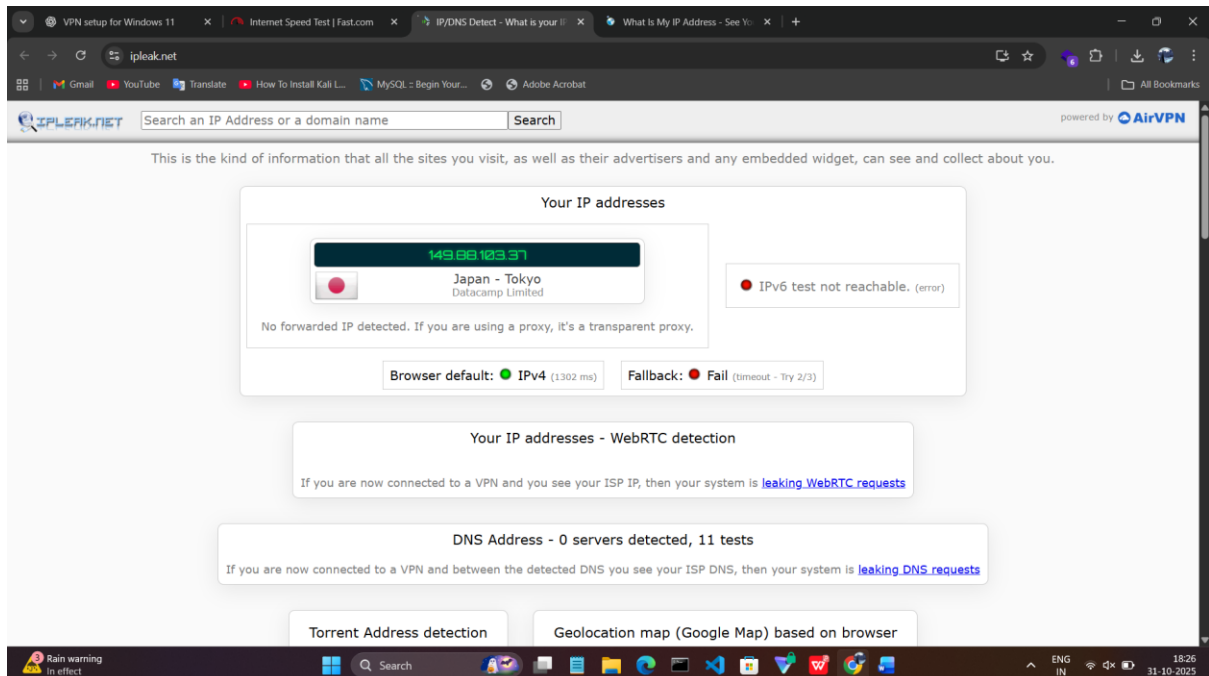


2. After Connecting VPN:

1. Click connect and wait until it shows “Connected”.



2. Check the new VPN IP and DNS leak test results to check whether the encryption of VPN tunnel is working.



VPN setup for Windows 11 | Internet Speed Test | Fast.com | IP/DNS Detect - What is your IP | What is My IP Address - See Your IP

ip leak.net

Torrent Address detection [Activate](#)

Geolocation map (Google Map) based on browser [Activate](#)
(may prompt a user permission on the browser)

IP Address details

IP:	149.88.103.37
ISP:	DataCamp Limited
AirVPN:	No
ASN:	212238
Country:	Japan (JP)
Region:	Tokyo (13)
City:	Tokyo
Time Zone:	Asia/Tokyo
Latitude & Longitude:	35.6893, 139.6899
Geolocation map (Google Map) based on IP Address	Activate
Accuracy Radius:	20 KM
Last data update:	Fri, 31 Oct 2025 12:56:28 +0000

Detected information

Your User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like)

System information (your browser, your language, your operating system, etc)

Platform: Win32

Screen information (your display hardware)

Your screen: 1536 x 864

Rain warning In effect

Search

ENG IN 18:28 31-10-2025

3. Check IP address and location.

VPN setup for Windows 11 | Internet Speed Test | Fast.com | IP/DNS Detect - What is your IP | What is My IP Address - See Your IP

whatismyipaddress.com

Enter Keywords or IP Address... [Search](#)

ABOUT | PRESS | PODCAST | SUPPORT

MY IP | IP LOOKUP | HIDE MY IP | VPNs | TOOLS | LEARN

My IP Address is:

IPv4: 149.88.103.37

IPv6: Not detected

My IP Information:

ISP:	DataCamp Limited
Services:	VPN, Server
City:	Tokyo
Region:	Tokyo
Country:	Japan

Looks like you're using a VPN! [RATE YOUR VPN](#)

[Show Complete IP Details](#)

Location not accurate? [Update My IP Location](#)

Japan 日本

Click for more details about 149.88.103.37

Which is your biggest concern about using the Internet?

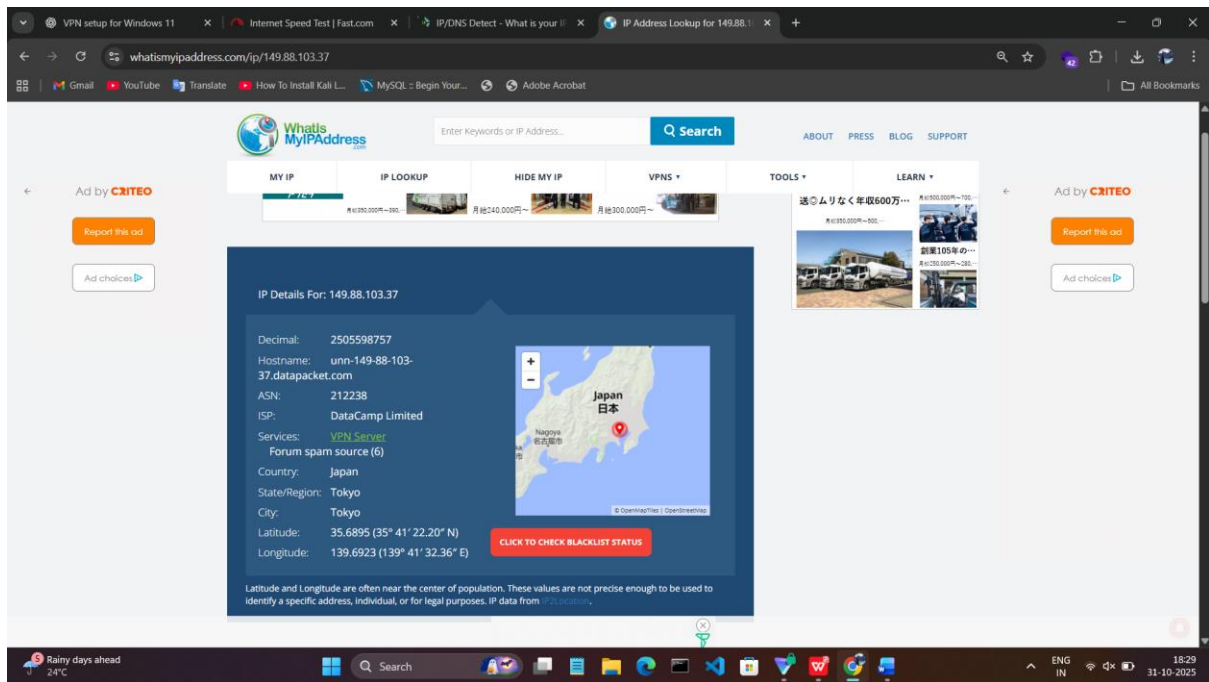
Security: I want protection from hackers on public Wi-Fi and other unsecured networks.

Privacy: I want to prevent my government, ISP and advertisers from tracking me.

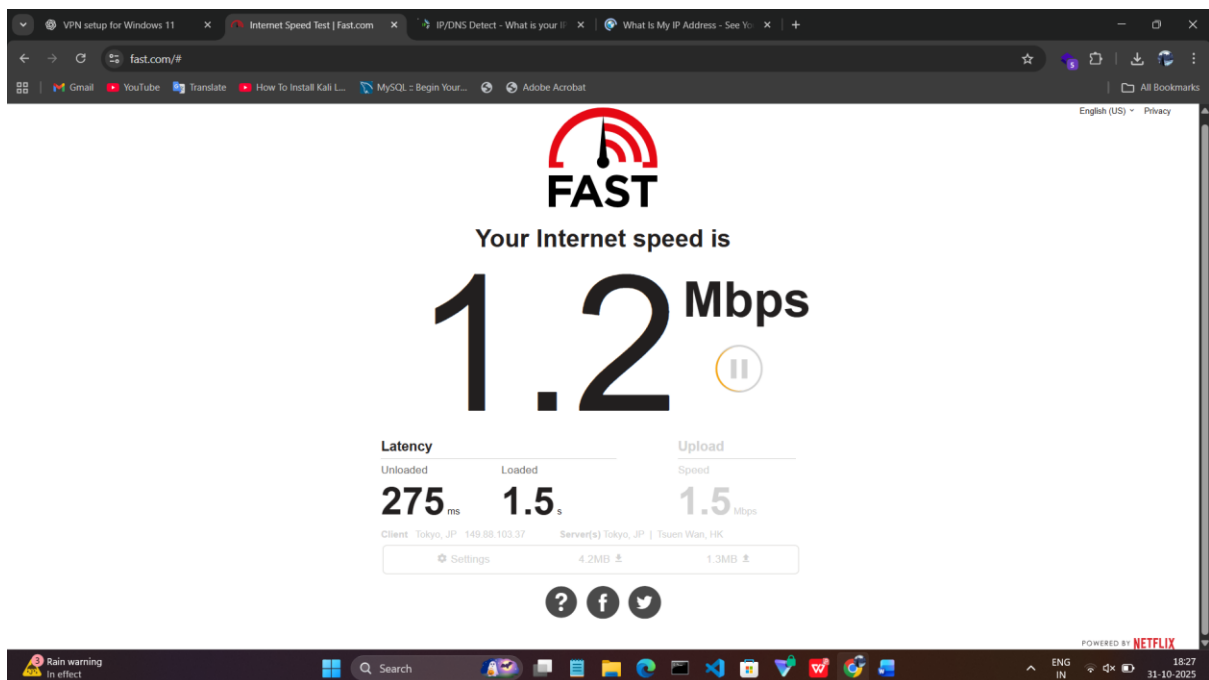
Access: I want to access Netflix, Hulu, and other streaming services without being blocked.

Top Stories IND v AUS: India...

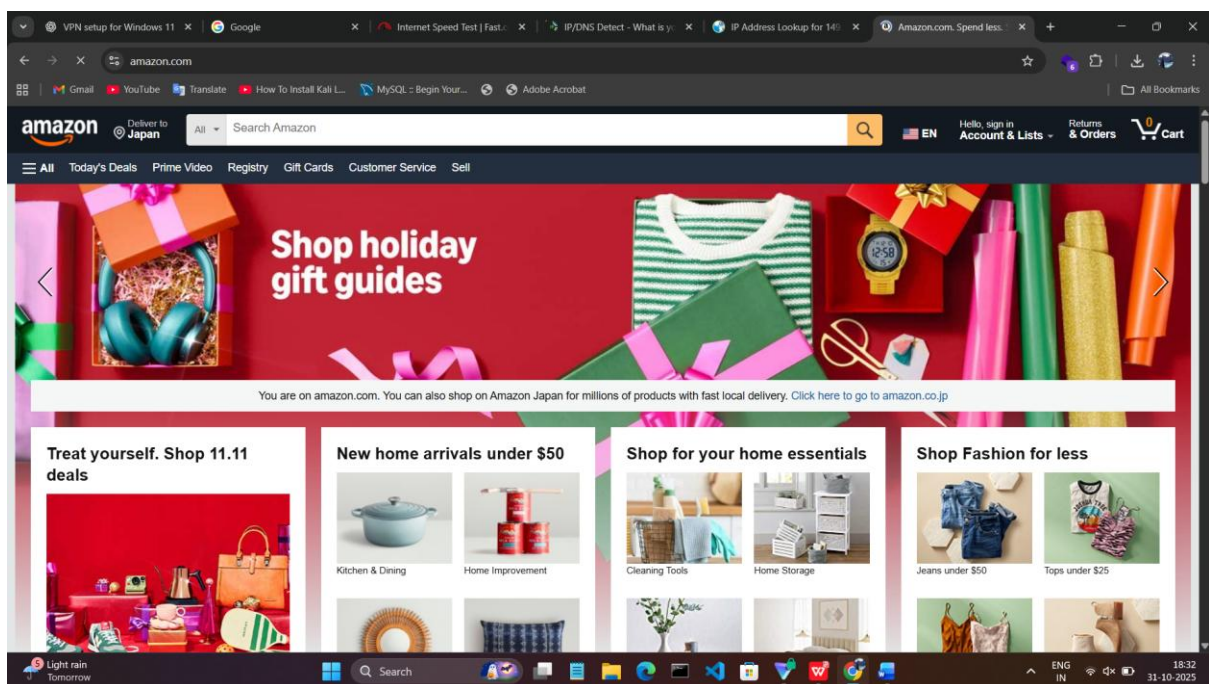
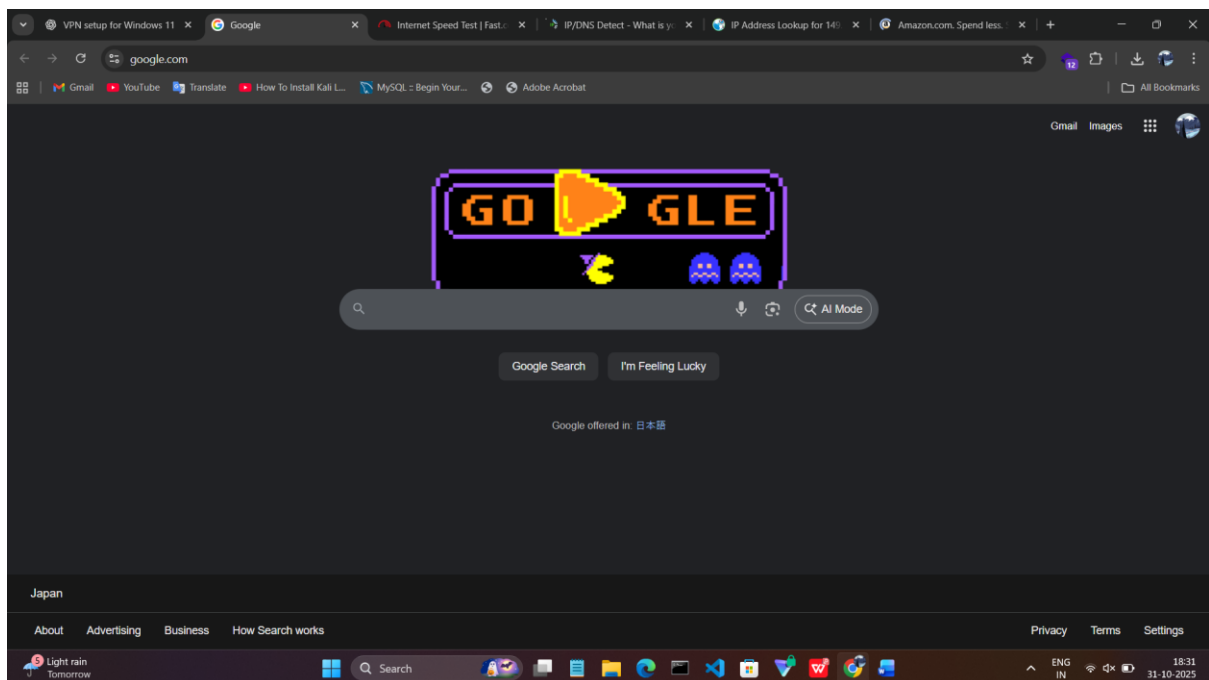
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4. Check browsing speed.

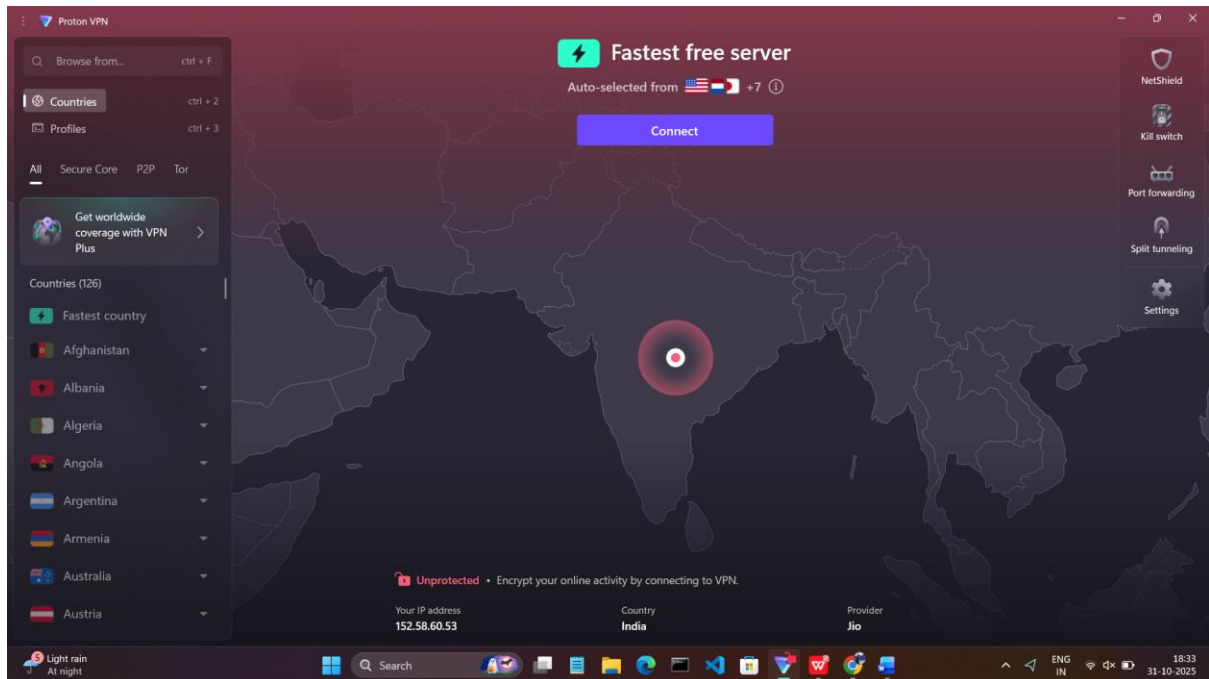


5. Visit websites to check the connection and speed.

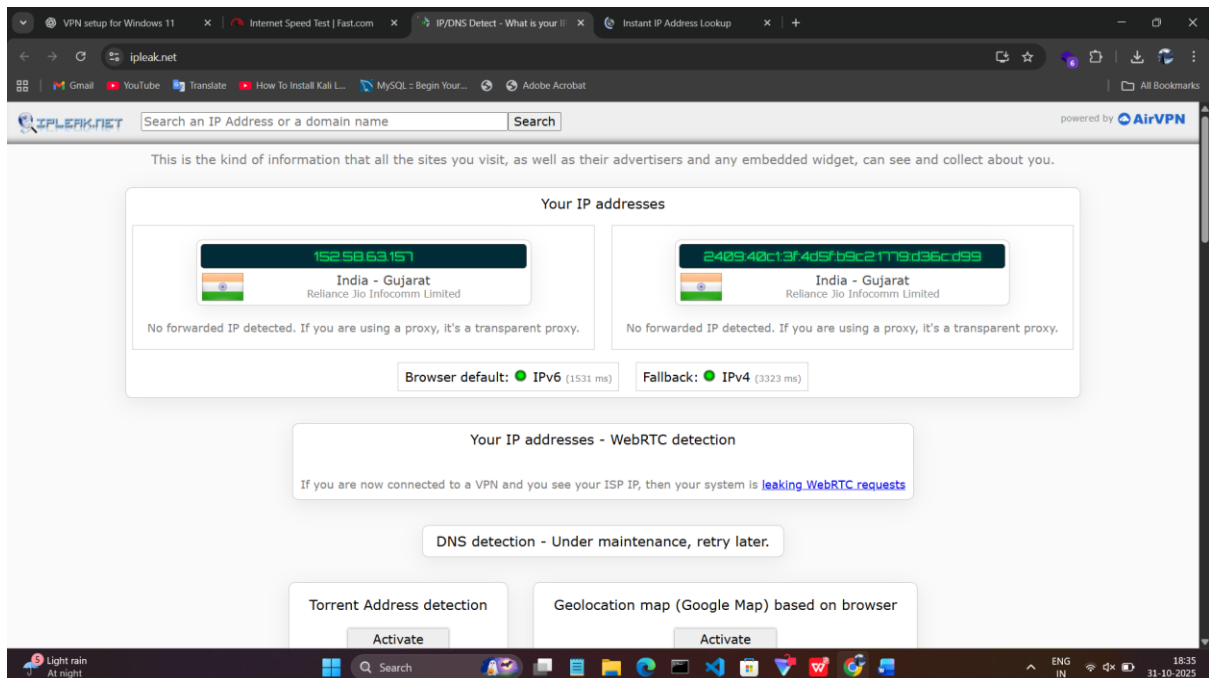


3. After Disconnecting VPN:

1. Then disconnect the VPN.



2. Check your new original IP and DNS leak test results showing.



3. See your location and IP Address.

The screenshot shows the 'What is My IP Address' website. The main content area displays the following information:

- My IP Address is:**
 - IPv6: 2409:40c1:3f:4d5f:b9c2:1779:d36c:d99
 - IPv4: 152.58.63.157
- My IP Information:**
 - ISP: Reliance Jio Infocomm Limited
 - City: Ahmedabad
 - Region: Gujarat
 - Country: India
- Your location may be exposed!** (with a red button: **HIDE MY IP ADDRESS NOW**)
- Show Complete IP Details** (link)
- Location not accurate?** (with a link: **Update My IP Location**)

A map of Ahmedabad, Gujarat, India, is shown next to the location information. The website also features a search bar at the top and a sidebar with links to 'ABOUT', 'PRESS', 'PODCAST', and 'SUPPORT'.

4. Check the internet speed.

The screenshot shows the 'Fast.com' website displaying internet speed test results. The main content area displays the following information:

- FAST** logo
- Your Internet speed is**
- 48 Mbps** (with a green circular icon)
- Latency**
 - Unloaded: 23 ms
 - Loaded: 968 ms
- Upload**
 - Speed: 1.7 Mbps
- Client:** Ahmedabad, IN | 2409:40c1:3f:4d5f:b9c2:1779:d36c:d99 | Jio
- Server(s):** Ahmedabad, IN | Navi Mumbai, IN | Jaipur, IN
- Settings** (button) | **180MB** (download icon) | **6.5MB** (upload icon)

The website also features a search bar at the top and a sidebar with links to 'ABOUT', 'PRESS', 'PODCAST', and 'SUPPORT'.

4. VPN Encryption and Privacy Features:

1. **Encryption Standard:**

ProtonVPN uses **AES-256-bit encryption**, which is a military-grade standard. This ensures that even if someone intercepts the traffic, they cannot decrypt or read it.

2. **VPN Protocols:**

It supports **OpenVPN**, **IKEv2/IPSec**, and **WireGuard** protocols. These create a secure “tunnel” for data transfer between your device and the VPN server.

3. **Secure Core Servers:**

ProtonVPN routes traffic through multiple servers (in privacy-friendly countries) before it exits to the internet, adding an extra layer of protection.

4. **DNS Leak Protection:**

During the test, the DNS leak check showed that all DNS queries went through the VPN — confirming no data leaks.

5. **Kill Switch & Always-On VPN:**

It has a built-in **Kill Switch**, which automatically blocks all internet traffic if the VPN disconnects unexpectedly — preventing IP exposure.

5. Benefits and Limitations of VPN:

➔ Benefits

1. Enhanced Privacy:

VPN hides your real IP address and location, making your online activities more private.

2. Data Encryption:

Encrypts all internet traffic, protecting sensitive data on public Wi-Fi or shared networks.

3. Bypass Restrictions:

Allows access to geo-blocked websites and content that might be unavailable in your region.

4. Protection from Tracking:

Prevents ISPs, advertisers, or hackers from tracking browsing activities.

5. Safe Public Wi-Fi Usage:

Provides a secure encrypted tunnel even on unsecured Wi-Fi networks (like cafés or airports).

➔ Limitations

1. Reduced Speed:

Internet speed may drop slightly due to encryption and routing through distant servers.

2. Free Tier Restrictions:

Free VPNs (like ProtonVPN's basic plan) have limited server locations and moderate speeds.

3. No Complete Anonymity:

A VPN hides your IP but doesn't make you completely anonymous online.

4. Connection Drops:

Temporary disconnections may expose your real IP unless the Kill Switch is enabled.

5. Potential Compatibility Issues:

Some streaming or banking sites block VPN traffic.